Yifan Xue

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Education

University of Pennsylvania — Philadelphia, PA
PhD Student in Mechanical Engineering — *Aug. 2022 – Current*

University of Southern California — Los Angeles, CA
Bachelor of Science in Mechanical Engineering, Minor in Computer Science — *Aug. 2018 – May 2022*

Publications

(**ITSC 2026**) N3P: Accelerated Automated Parking via a Learning-Based Naturalistic Three-Stage Scheme [Paper]

(**RA-L 2026**) Proactive Local-Minimum-Free Mobile Robot Navigation using MMP-MCBF Framework [Paper] [Video]

(**SII 2026**) RNBF: Real-Time RGB-D Based Neural Barrier Functions for Safe Robotic Navigation [Paper]

(**ICRA 2025**) ADMM-MCBF-LCA: A Layered Control Architecture for Safe Real-Time Navigation [Paper] [Video]

(**IROS 2025**) Gradient Field-Based Dynamic Window Approach for Collision Avoidance in Complex Environments [Paper]

(**Under Review**) Memory-Aware Multi-Sensor Perception for Efficient and Safe Navigation in Dynamic Environments [Paper] [Video]

(**Under Review**) No Minima, No Collisions: Combining Modulation and Control Barrier Function Strategies for Feasible Dynamic Collision Avoidance [Paper] [Video]

(**ICRA Workshop 2025**) Modulated Control Barrier Function for Freezing Problem in Social Navigation [Paper]

Research

Learning Bi-directional Communication for Robot Safety and Liveness — Aug 2026 – Present
University of Pennsylvania — *Philadelphia, PA*

- Developing a bi-directional layered architecture where low-level safety and feasibility feedback guides high-level policy adaptation, beyond conventional one-way planning.
- Addressing conflicts between high-level actions and low-level safety filters that can disrupt motion, cause infeasibility, or distort actions enough to prevent task completion.
- Training reinforcement learning modules for low-level liveness enhancement and high-level policy adaptation using safety-filter feedback.
- Will validate the learning pipeline on mobile manipulators through competitive sports tasks with human players.

Memory-Aware Multi-Sensor Perception for Efficient and Safe Navigation — Jan 2025 – Aug 2026
University of Pennsylvania — *Philadelphia, PA*

- Developed an online multi-sensor perception and environment learning and modeling framework that fuses 360° LiDAR and RGB-D camera data for autonomous navigation in previously unseen environments.
- Conducted hardware experiments and Gazebo simulations to evaluate learning-enabled navigation in complex indoor and outdoor environments without relying on motion-capture systems.
- Best Paper Finalist at the IROS Workshop on Long-Term Perception for Human-Centric Autonomy
- Research under review at IEEE International Conference on Robotics and Automation (ICRA 2027).

Proactive Mobile Robot Navigation using Multimodal Motion Prediction — Aug 2024 – May 2025
University of Pennsylvania — *Philadelphia, PA*

- Implemented a CNN-based Energy-Based Model (EBM) for multimodal trajectory prediction, learning probabilistic distributions over future motions of dynamic agents.
- Integrated learned multimodal motion prediction with optimization-based planning to generate proactive collision-free trajectories in complex, crowded, dynamic environments.

Modulated Control Barrier Functions for Dynamic Collision Avoidance May 2023 – May 2025
University of Pennsylvania *Philadelphia, PA*

- Analyzed theoretically, qualitatively, and quantitatively the connections and differences between the reactive safe control methods of Dynamical Systems Modulation (Mod-DS) and Control Barrier Function Quadratic Programs (CBF-QP).
- Proposed Modulated CBF-QP (MCBF-QP), a novel safe control method that combines the local-minima-free strength of Mod-DS with CBF-QP's generalizability to control-affine systems.
- Accelerated the proposed MCBF-QP implementation using JAX through just-in-time (JIT) compilation and vectorized computation, enabling real-time execution.

ADMM Layered Control Architecture for Safe Real-Time Navigation Aug 2023 – Sept 2024
University of Pennsylvania *Philadelphia, PA*

- Proposed a Layered Control Architecture (LCA) integrating an ADMM-based nominal planner with a CBF-QP safety filter for safe navigation in dynamic environments with moving obstacles and input saturation constraints.
- Developed an intelligent real-time path selection mechanism that selects trajectories from an offline path library based on current obstacle positions and velocities, and enables dynamically-feasible path tracing and switching for nonholonomic robots via online evaluation of precomputed path adapter functions.

Working Experience

Honda Research Institute | *Autonomous Driving Research Intern* Summer 2025

- Proposed novel local parking path planning algorithms by integrating machine learning models (KNN, MLP) with Hybrid A*, achieving faster path computation times than existing methods in autonomous parking scenarios.
- Implemented and trained, in Pytorch, a Transformer-based RL agent with multi-head self-attention for autonomous parking, serving as a baseline to compare against the proposed method.

University of Pennsylvania | *MEAM6230 Head Teaching Assistant* Spring 2026

- *MEAM6230: Learning and Control for Adaptive and Reactive Robots* introduces the fundamental principles of the modern Dynamical Systems (DS) paradigm for motion planning, learning, and control.
- Designed student homework assignments on obstacle avoidance using on-Manifold Modulation.

University of Pennsylvania | *MEAM5200 Head Teaching Assistant* Fall 2023 - Spring 2025

- *MEAM5200: Introduction to Robotics* is a graduate-level introductory course to manipulation kinematics, motion planning and control.
- Held lab sessions and taught students to use Panda Arm.
- Designed Panda-control-related student homework assignments on obstacle avoidance.

Awards

Center for Undergraduate Research in Viterbi Engineering (CURVE) Fellowship Spring 2021 – Summer 2021
Winners of the 2021 CURVE Symposium May 18, 2021
USC Provost Fellowship Spring 2021 – Spring 2022
USC Academic Achievement Award (AAA) Fall 2019 – Spring 2022

Technical Skills

Programming: Python, C++, MATLAB
Control: Trajectory optimization, control theory, graph-based path planning, MPC, CBF, ADMM
Machine Learning: Reinforcement learning, neural networks, transformer architecture, Pytorch, JAX
Robot perception: LiDAR, RGB-D camera
Computer-Aided Design: Solidworks, NX Nastran, Fusion 360